	ℓ	**Top-scoring tokens by the subject representation**
GPT-2	25	Jr, Sr, era, Era, MP, JR, senior, Clarence, final, stars, Junior, High, Architects
	35	Hockey, hockey, Jr, NHL, Rangers, Canucks, Sr, Islanders, Leafs, Montreal, Oilers
	40	NHL, Hockey, Jr, Canucks, retired, hockey, Toronto, Sr, Islanders, Leafs, jersey, Oilers
GPT-J	13	Jr, Archives, ice, ring, International, Jersey, age, Institute, National, aged, career
	17	hockey, NHL, Jr, Hockey, ice, Canadian, Canada, Archives, ice, Toronto, ring, National
	22	NHL, National, hockey, Jr, Foundation, Hockey, Archives, Award, ice, career, http

Table 3: Top-scoring tokens by the subject representations of "Mark Messier" across intermediate-upper layers (ℓ) in GPT-2 and GPT-J.

	ℓ	**Top-scoring tokens by the subject representation**
GPT-2	25	Edition, Ribbon, Series, Mini, version, Card, CR, XL, MX, XT, Cube, RX, Glow, speakers
	35	iPod, Bluetooth, Apple, Mini, iPhone, iOS, speaker, Edition, Android, Controller
	40	iPod, headphone, Bluetooth, speakers, speaker, Apple, iPhone, iOS, audio, headphones
GPT-J	13	Series, device, Apple, iPod, model, Music, iPhone, devices, models, style, music
	17	Series, model, device, style, Music, series, Archives, music, design, interface, models
	22	iPod, Music, music, song, Series, iPhone, Apple, songs, Music, review, series, Audio

Table 4: Top-scoring tokens by the subject representations of "iPod Classic" across intermediate-upper layers (ℓ) in GPT-2 and GPT-J.

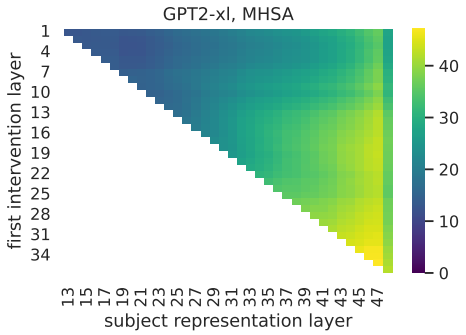

Figure 16: The attributes rate of the subject representation at different layers, when intervening on (canceling) different 10 consecutive MHSA sublayers in GPT-2.

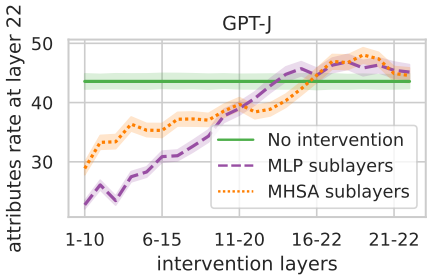

Figure 17: The attributes rate of the subject representation with and without canceling updates from the MLP and MHSA sublayers in GPT-J.

shows the effect of replacing intermediate representations with early layer representations at different positions on the attribute extraction rate (§7).

Overall, these results are consistent with those for GPT-2 described throughout the paper.

F Analysis of MLP Outputs

Following the observation that the early MLP sublayers are crucial for attribute enrichment of subject representations, we further analyze their updates to these representations. To this end, we decompose these updates into sub-updates that, according to Geva et al. (2022b), often encode human-interpretable concepts. Concretely, We decompose Eq. 6 to a linear combination of parameter vectors of the second MLP matrix:

$$\mathbf{m}_i^\ell = \sum_{j=1}^{d_{inner}} \sigma\left(\mathbf{w}_I^{(\ell,j)}\left(\mathbf{a}_i^\ell + \mathbf{x}_i^{\ell-1}\right)\right) \cdot \mathbf{w}_F^{(\ell,j)},$$

where d_{inner} is the MLP inner dimension, and $\mathbf{w}_I^{(\ell,j)}, \mathbf{w}_F^{(\ell,j)}$ are the j-th row and column of W_I^ℓ, W_F^ℓ, respectively. We then take the 100 vectors in W_F^ℓ with the highest contribution to this sum, for $\ell = 1, ..., 20$ in GPT-2, and inspect the top-scoring tokens in their projections to E.

From manual inspection, we indeed were able to identify cases where concepts related to the input subject are promoted by the dominant sub-updates. Examples are provided in Tab. 6. We note that

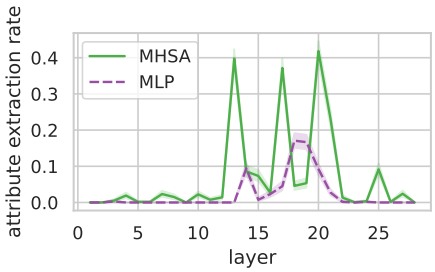

Figure 18: Attribute extraction rate across layers, for the MHSA and MLP sublayers in GPT-J.

	Extraction rate	# of extracting layers
MHSA	76.7	1.9
- last	46.7	0.8
- non-subj.	45.6	0.8
- all non-subj. but last	44.4	0.9
- subj. last	41.1	0.7
- subj. last + last	40.3	0.7
- all but subj. last + last	34.8	0.5
- all but subj. last	31.3	0.5
- subj.	27.8	0.3
- all but last	24.8	0.3
- all but first	0	0
MLP	41	0.6

Table 5: Per-example extraction statistics across layers, for the MHSA and MLP sublayers, and MHSA with interventions on positions: (non-)subj. for (non-)subject positions, last (first) for the last (first) input position.

quantifying this process is non-trivial (either with human annotations or automatic methods), and so we leave this for future work.

G Subject-Attribute Mappings in Attention Heads

In §7, we showed that for many extraction events, it is possible to identify specific heads that encode mappings between the input subject and predicted attribute in their parameters. We provide examples for such mappings in Tab. 7.

As a further analysis, we inspected the heads that repeatedly extract the attribute for different input queries in GPT-2. Specifically, for the matrix $W_{VO}^{\ell,j}$ of the j-th head at the ℓ-th layer, we analyze its projection to the vocabulary $G^{\ell,j}$ (Eq. 8), but instead of looking at the top mappings for a specific subject-token (i.e. a row in $G^{\ell,j}$), we look at the top mappings across all rows, as done by (Dar et al., 2022). In our analysis, we focused on the heads that extract the attribute for at least 10% of the queries (7 heads in total). We observed that these heads act as "knowledge hubs", with their

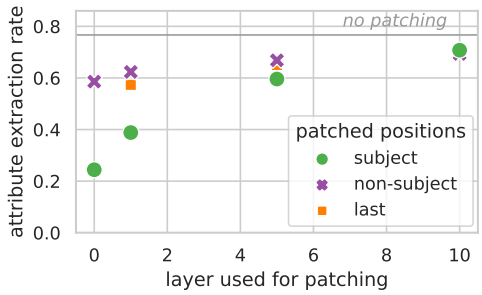

Figure 19: Extraction rate when patching representations from early layers at different positions in GPT-J.

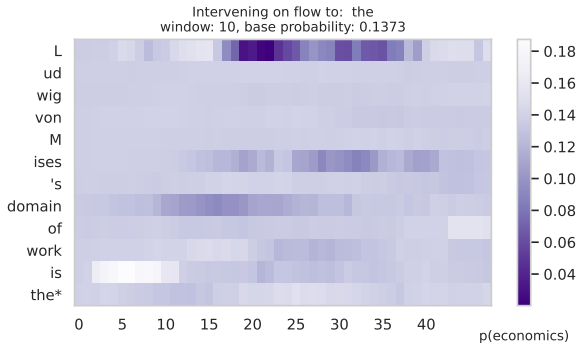

Figure 20: Example intervention on information flow to the last position of the input `Ludwig von Mises's domain of work is the`, using a window size 10, in GPT-2.

top mappings encoding hundreds of factual associations that cover various relations. Tab. 8 shows example mappings by these heads.

H Example Interventions on Information Flow

Example interventions in GPT-2 are shown in Fig. 20, 21, 22, 23, 24, and in GPT-J in Fig. 25, 26, 27, 28, 29.

Subject	Description	Layer, Dim.	Top tokens in the projection
United Launch Alliance	American spacecraft launch service provider	1, 5394	`jet, flights, aircraft, Cargo, passenger, plane, rockets, airspace, carrier, missiles, aerospace`
Yakuza 2	Action-adventure video game	2, 4817	`Arcade, pixels, arcade, Wii, Fighters, pixels, Minecraft, Sega, Sonic, GPUs, Hardware, downloadable, multiplayer, Developers, livestream`
Leonard Bernstein	American composer, pianist, music educator, and author	4, 248	`violin, rehearsal, opera, pian, composer, Harmony, ensemble, Melody, piano, musicians, poets, Orchestra, Symphony`
Brett Hull	Canadian–American former ice hockey player	4, 5536	`discipl, athlet, Athletics, League, Hockey, ESPN, former, Sports, NHL, athleticism, hockey, Champions`
Bhaktisiddhanta Saraswati	Spiritual master, instructor, and revivalist in early 20th century India	5, 3591	`Pradesh, Punjab, Bihar, Gandhi, Laksh, Hindu, Hindi, Indian, guru, India, Tamil, Guru, Krishna, Bengal`
Mark Walters	English former professional footballer	10, 1078	`scorer, offence, scoring, defences, backfield, midfielder, midfield, striker, fielder, playoffs, rebounds, rushing, touchdowns`
Acura ILX	Compact car	19, 179	`Highway, bike, truck, automobile, Bicycle, motorists, cycle, Motor, freeway, vehicle, commute, Route, cars, motorcycle, Tire, streetcar, traffic`
Beats Music	Online music streaming service owned by Apple	20, 5488	`Technologies, Technology, Apple, iPod, iOS, cloud, Appl, engineering, software, platform, proprietary`

Table 6: Top-scoring tokens in the projection of dominant MLP sub-updates to the subject representation in GPT-2.

Matrix	Subject	Top mappings
$W_{OV}^{(33,13)}$	*Barry Zito*	`Baseball, MLB, infield, Bethesda, RPGs, pitching, outfield, Iw,` **`pitcher`**
$W_{OV}^{(37,3)}$	*iPod Classic*	`Macintosh, iPod, Mac, Perse,` **`Apple`**`, MacBook, Beck, Philipp`
$W_{OV}^{(44,16)}$	*Philippus van Limborch*	`Dutch, Netherlands, van,` **`Amsterdam`**`, Holland, Vanessa`

Table 7: Example mappings between subject-tokens (underlined) and attributes (**bold**) encoded in attention parameters (tokens for the same word are not shown).

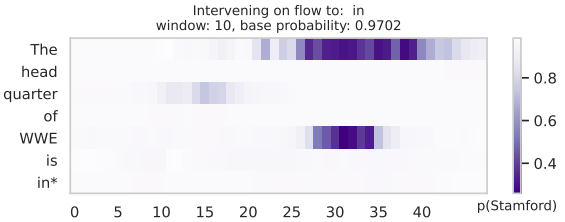

Figure 21: Example intervention on information flow to the last position of the input `The headquarter of WWE is in`, using a window size 10, in GPT-2.

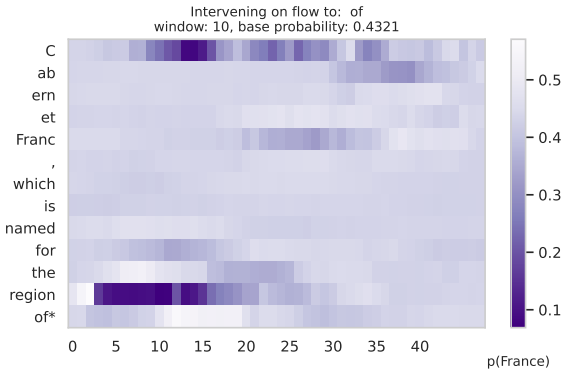

Figure 22: Example intervention on information flow to the last position of the input `Cabernet Franc, which is named for the region of`, using a window size 10, in GPT-2.